

OS LAB MANUAL (CS23431)

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File Organization Technique- Single and Two level directory

Aim: To implement File Organization Structures in C are:

- a. Single Level Directory
 - b. Two-Level Directory
-

A) Single Level Directory

Program:

```
#include <stdio.h>
#include <string.h>
int main() {
    int n, i;
    char files[10][30];
    printf("Enter the number of files: ");
    scanf("%d", &n);
    printf("\nCreating Single-Level Directory Structure...\n");
    for (i = 0; i < n; i++) {
        printf("\nEnter the name of file %d: ", i + 1);
        scanf("%s", files[i]);
        printf("\n+-----+\n");
        printf(" | Root Directory | \n");
        printf("+-----+\n");
    }
}
```

```

        for (int j = 0; j <= i; j++) {
            printf("        |\n");
            printf("        +--> [ %s ]\n", files[j]);
        }
    }
    return 0;
}

```

Input:

```

Enter the number of files: 2
Creating Single-Level Directory Structure..
Enter the name of file 1: A
Enter the name of file 2: D

```

Output:

```

Enter the number of files: 2
Creating Single-Level Directory Structure...
Enter the name of file 1: A
+-----+
| Root Directory |
+-----+
      |
      +--> [ A ]
Enter the name of file 2: D
+-----+
| Root Directory |
+-----+
      |
      +--> [ A ]
      |
      +--> [ D ]

```

B) Two-Level Directory :

Program:

```
#include <stdio.h>

#include <string.h>

struct File {

    char name[30];

};

struct User {

    char name[30];

    int fileCount;

    struct File files[10];

};

int main() {

    int userCount, i, j;

    struct User users[10];

    printf("Enter the number of users (directories): ");

    scanf("%d", &userCount);

    for (i = 0; i < userCount; i++) {

        printf("\nEnter the name of User %d: ", i + 1);

        scanf("%s", users[i].name);

        printf("Enter number of files for %s: ", users[i].name);

        scanf("%d", &users[i].fileCount);

        for (j = 0; j < users[i].fileCount; j++) {

            printf("Enter file %d name for %s: ", j + 1, users[i].name);

            scanf("%s", users[i].files[j].name);

        }

    }

    printf("\n\nTwo-Level Directory Structure:\n");

    printf("+-----+\n");

    printf("|    Root Directory    |\n");

    printf("+-----+\n");
```

```

for (i = 0; i < userCount; i++) {

    printf("    |\n");

    printf("    +--> User: %s\n", users[i].name);

    for (j = 0; j < users[i].fileCount; j++) {

        printf("        |\n");

        printf("        +--> File: %s\n", users[i].files[j].name);

    }

}

return 0;
}

```

Input:

```

Enter the number of users (directories): 2

Enter the name of User 1: joe
Enter number of files for joe: 2
Enter file 1 name for joe: A
Enter file 2 name for joe: B

Enter the name of User 2: Roy
Enter number of files for Roy: 2
Enter file 1 name for Roy: A
Enter file 2 name for Roy: B

```

Output:

```

Enter the number of users (directories): 2

Enter the name of User 1: joe
Enter number of files for joe: 2
Enter file 1 name for joe: A
Enter file 2 name for joe: B

Enter the name of User 2: Roy
Enter number of files for Roy: 2
Enter file 1 name for Roy: A
Enter file 2 name for Roy: B

Two-Level Directory Structure:
+-----+
|      Root Directory      |
+-----+
|
| +--> User: joe
|   |
|   | +--> File: A
|   |   |
|   |   | +--> File: B
|   |
| +--> User: Roy
|   |
|   | +--> File: A
|   |   |
|   |   | +--> File: B
|

```